Text Analysis for IMBD TV Show Ratings and Reviews

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Introduction

Tv shows are a beloved form of entertainment across the world. People from different countries and who speak different languages tune into their favorite shows daily. With the diverse population of TV show watchers, diverse opinions are present too. Not everyone likes the same show, and they make that apparent in the reviews they leave.

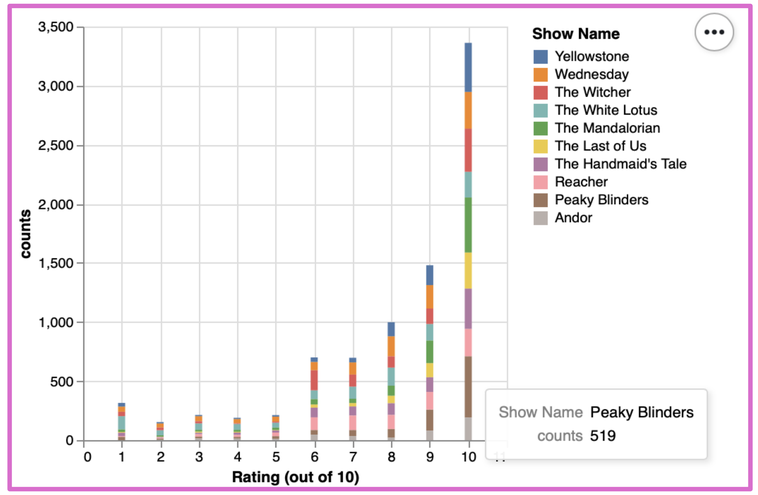
For this project we took a large dataset from IMBD and analyzed the reviews of 10 TV shows: Andor, Peaky Blinders, Reacher, The Handmaid's Tale, The Last of Us, The Mandalorian, The White Lotus, The Witcher, Wednesday, and Yellowstone. Using Sentiment Analysis, Topic Modeling, and Descriptive Analysis and Visualization using Altair, we analyzed the number and context of reviews.
Rationale: Sentiment Analysis was chosen due to its ability to understand the reviewer's emotions in the review. Topic Modeling was used to understand what the reviews were mainly about. With the use of Descriptive Analysis, simplifying the large dataset and identifying trends and patterns was the goal. In addition to these methods, we built a machine learning model with AdaBoost as our classifier. Graphics were created to visualize the findings.

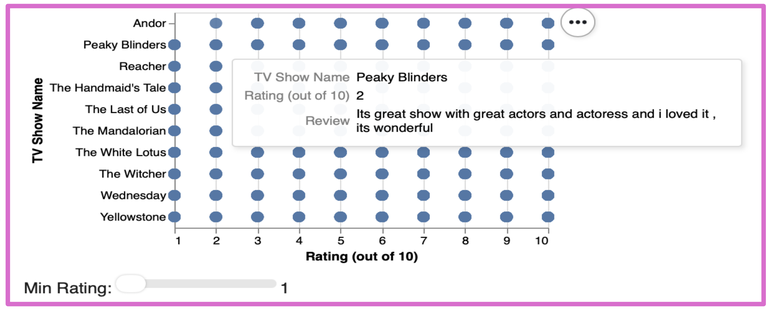
Data

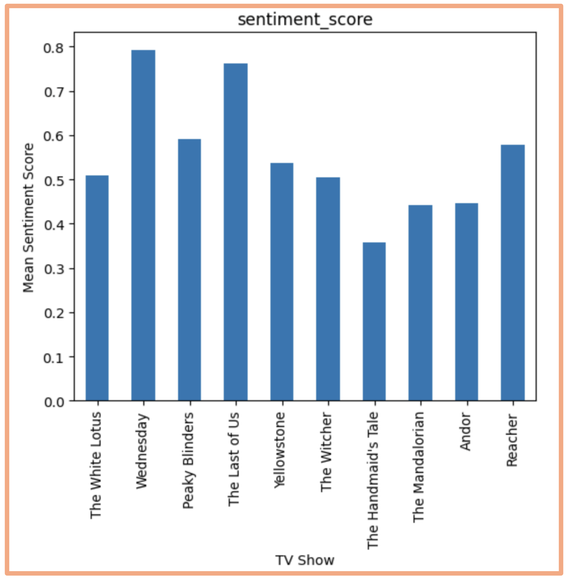
- Extracted from Kaggle under search "TV Show Reviews"
- Dataset includes: Review IDs, Show IDs, Ratings (out of 10), and Reviews
- Last date updated: June 2025
- Total entries: 8318

Methodology

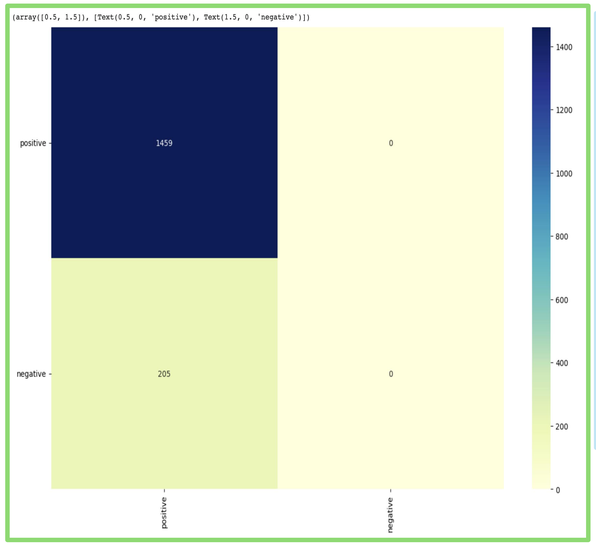
- Performed Sentiment Analysis and constricted a graph showing the average tone for each TV show included in the dataset
- Created a Topic Model visualizing the most common words in 20 topics
- Built a machine learning model using Sklearn
- For the machine model, split the dataset randomly 80/20 between training data and testing data
- Applied AdaBoost classifier model to predict and classify reviews as negative (rating less than or equal to 5) or positive (rating greater than 5).

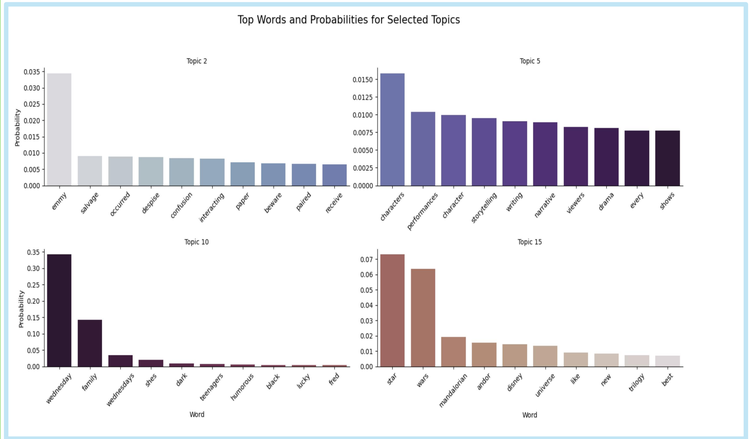






	sentiment_score
count	8319.000000
mean	0.555964
std	0.616758
min	-0.998300
25%	0.408000
50%	0.881800
75%	0.968600
max	0.999700





Dataset

GitHub

Findings

General Sentiment: On average people felt neutral about all 10 TV shows (.56 sentiment score)
Highest Sentiments: People spoke the most positively about the show Wednesday
Lowest Sentiments: People spoke the most negatively about The Handmaid's Tale
Most 10 Ratings: Peaky Blinders (519)
Most 1 Ratings: The White Lotus (113)
Most Frequent Words: Correlated with content of show
Note: Not all ratings match the attitude of the review given

Model Performance

- Predicted review sentiments on the testing dataset with an accuracy and precision rate of about 88%
- Model could not predict negative review ratings due to data being heavily skewed towards high review ratings

Conclusion

Using Sentiment Analysis, Topic Modeling, Descriptive Analysis, and Supervised Machine Learning, we examined the emotional tone of TV show reviews and how they related to their consequent review rating. We found that the average tone of every review was neutral (5/10). The review ratings and learning machine revealed that the dataset was too heavily skewed towards high scores and sometimes inaccurate compared to the review left. This made getting accurate results and predictions reliant on Sentiment Analysis. In the future, going a step further to create a model that predicts ratings from the sentiment of reviews would be recommended.

References

Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). Prediction. In *Text as data: A new framework for machine learning and the social sciences*. Princeton University Press.

Lamba, M., & Madhusudhan, M. (2022). Information visualization. In *Text mining for professionals: An uncharted territory* (pp. 243–288). Springer. https://doi.org/10.1007/978-3-030-85085-2_7

Lamba, M., & Madhusudhan, M. (2022). Sentiment analysis. In *Text mining for information professionals: An uncharted territory* (pp. 191–211). Springer. https://doi.org/10.1007/978-3-030-85085-2_7

Lamba, M., & Madhusudhan, M. (2022). Topic modeling. In *Text mining for information professionals: An uncharted territory* (pp. 105–136). Springer. https://doi.org/10.1007/978-3-030-85085-2_7

Vinayak Soni (n.d.). *IMDB TV show reviews dataset* [Data set]. Kaggle. https://www.kaggle.com/datasets/vinayaks0n1/imdb-tv-show-reviews